Carnegie Mellon University
Computer Science Academy

Design and Development Guide

Before beginning this project, be sure to review the rubric to ensure you include all the required elements.

Description

Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

I plan to make an interactive game mimicking a person's activities within a day of time, the day starts by shooting and catching wildlife, and ends by cooking them for dinner. I want to make a pretty scene from the window that feels dreamy yet unreal; it will likely contain macaron colors or bright, sharp colors to make the scene look artificial. The scene will probably be a green field with animals, such as rabbits, followed by a flowing river and mountains that fade into the blue sky. I envisage a sun in the right corner, while it gives off rays of light onto the scene. Moreover, I wish there could be an interaction mechanism where the user has to press the mouse to open the curtains and reveal the scene themself. There will probably be a few birds in the sky doing a diagonal wraparound motion. I wish to turn this into a game, where the user can use their mouse to "kill the birds" by pressing on them. After the bird is shot, the user can pick up the bird for dinner. The animals in the grass field and the water can also be grabbed for dinner. I want to also add a navigation. Bar on the right top to show the "supplies" the user has for dinner. After grabbing all the "supplies", the user may go home and cook them. I envision it as the user puts all the supplies into one pot and opens the fire to boil them. After around 10 seconds, the meal is ready, and the user's "day in the game" is over, with the user eating what he caught during the "day".

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?

(0,0)

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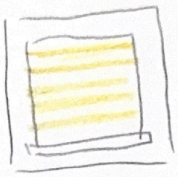
Colors:
- rgb(255,229,236)
- rgb(234,246,250)
- rgb(250,234,185)
- rgb(2,163,250)
- rgb(117,77,8)
- rgb(237,231,40)
- grey
- rgb(134,212,131)
- rgb(25,74,122)
- rgb(33,130,129)
- rgb(63,176,58)
- rgb(255,183,76)
- rgb(255,124,28)
- white
- rgb(255,143,174)
- black

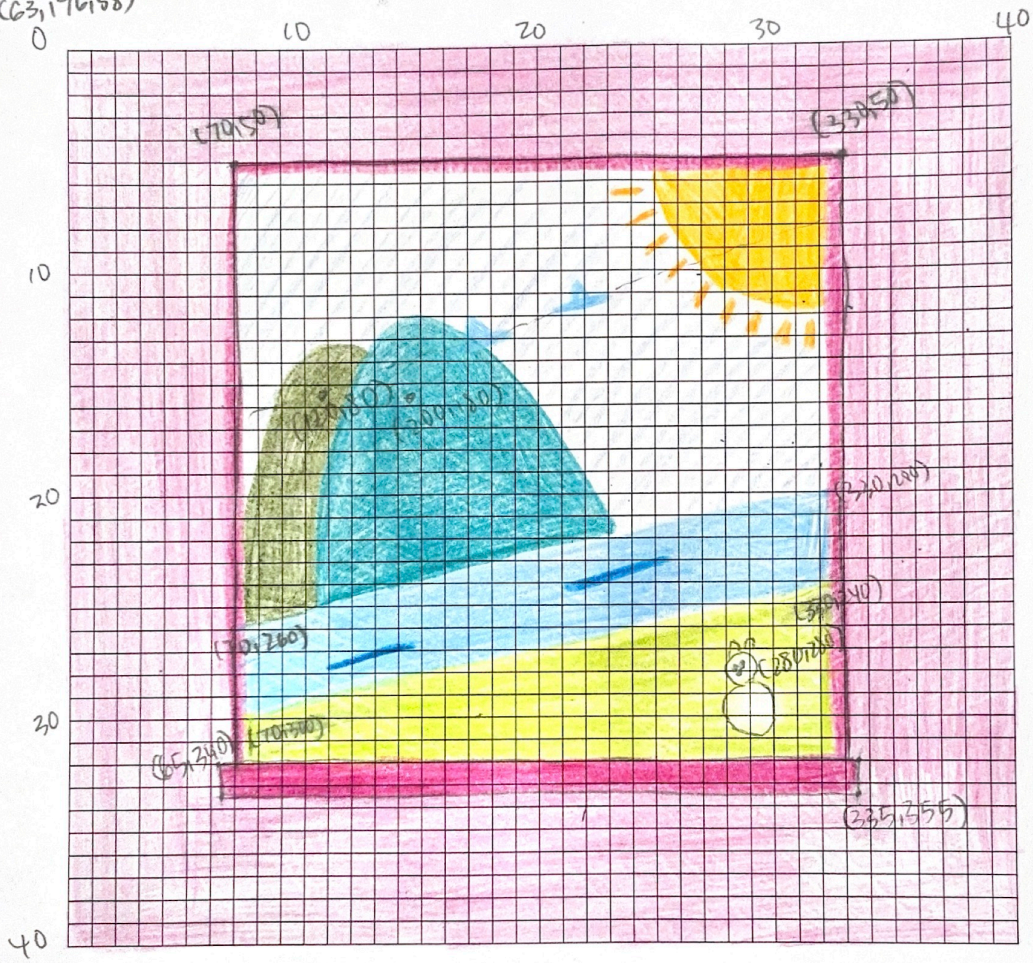
3 Different scenes!

Curtain

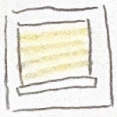


Outdoor

Indoor



Game interaction + Flow.

① 

onMousePress ⇒ curtain up

②

Sun + River lines are animations.

onMousePress clickonbirds x2 ⇒ Birds on Grass (Die)

③

Birds "Die"

onMousePress clickonbirds x2 ⇒ BirdCount +2 Birddissapear

④

Bird dissapear

on mousepre clickonrabb ⇒ rabbit count + rabbit dissape

⑤

Birdcount x2
Rabbitcount x1

onStep fade dark ⇒ Automatic

⑥

dark += 5
light += 5

onMousepress light ⇒ Go from outdoor to indoor

⑦

Food cooking
20 steps

onStep 20 steps ⇒

⑧

⇒ appStop !!

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Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
Shape Objects / Gradients and Colors	Shapes are always something needed in a program, and their gradient and color choices will add an aesthetic touch to the scene as well. I need them to build up my scene and story.
Variables	I plan to use variables in my code to make it easier to attribute them in my future functions. With the use of Groups, we also need a variable to name them into one shape and control the shapes as a whole. For instance, if I want a bird to disappear from the sky, I can simply say: "bird.visible = False" instead of listing out all its shapes and making them disappear.
Properties	Properties will be very helpful when it comes to complex animations, such as the visibility of the birds will change after they are shot, or the background will fade into the darkness gradually by changing its opacity.
Groups (NEW)	Groups are something we recently learned; they come in handy when you want to program movements of complex shapes. In my case, it will be the animals that need to be grouped, as I wish for the multiple shapes to move together, such as the birds that are made of 2 shapes to fly across the sky as one.
Event Functions (NEW)	I plan to use the "onMousePress" and "onStep" functions to animate and allow interactions with the user. "onStep" is something me recenly leanred, and it allows for object motion even without the user doing anything. My animals, river, and sun will probably be moving due to this function. On the other hand, "onMousePress" requires the user to press the screen; I will probably use this function for interactions like opening the curtains or shooting the animals.
Conditionals	Conditionals will occur very often throughout my code. Testing statements like if or else will appear to push forward the whole story and scene changes.

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?

- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

Throughout the creation of this scene, I absolutely saw an opportunity to make the scene look more animated and thoughtful. While programming, I was unsure of how to make the scene change from outdoor to indoor seem less abrupt, while I also wanted to incorporate something that we learned using Unit 6; therefore, I thought of using "onStep" to make the background gradually change into indoors by changing the opacity of a black block in front of the page. This makes the scene seem as if the day ended as things darken, then after the whole page becomes black, a light switch is visible, and after the user presses it in order to turn on the lights of the room. This transition of fading from outdoor to opening the lights of your "kitchen" is a smooth transition detail that I have not envisaged before. This detail made my whole program seem much smoother and logical, while it did not change much of the aspects of my code. The light switch was hard to turn on, as if its visibility was simply tuned to "false"; then the user may accidentally press on it while it was outdoors, shooting birds. Therefore, I had to add another condition to make it appear only when the background is black; thinking of this took me a lot of time. I originally envisioned the transition as abruptly changing into indoors, but I later replaced the transitoon into this.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
Variables	87-90	Variables are names that store an object or certain values that can be used later in the program.	Through using variables, I can come to them later in my event functions and change their appearance. For instance, the lables visiblity change all the time through my code, and with variable attributes to them, they make changing their appearance much easier.

Shape Objects/Colors	6-21	Shapes can be created by entering the parameters of the shape using a set form, such as for a circle, you give (centerX, centerY, radius, others). Adding colors and gradients to a shape that is initially designed to be black.	Shapes and objects make up my code; I build my background and scene with them. I also picked colors from the RGB scale to make the picture look aesthetic and visually appealing. The shapes can also change if you tune their properties.
Properties	123 - 129	Properties are attributes that change the shape's appearance and location, such as setting values for visibility and center location.	Through changing properties, I can change the visual appearance of things to allow the animation to continue with logic, such as the visibility, height, width, and center of the object to make it as if it is moving, changing position, or disappearing. For instance, the rabbit is being "caught" by the user, so it disappears, while a label appears to indicate that the user has caught the rabbit.
Groups (NEW)	53 - 59	Combining a series of shapes and elements through "group" using brackets allows control for multiple shapes at once.	I used the grouping concept to utilize my space and organize the program. For example, when constructing birds that fly in the air, I used up to 2 shapes that, on their own, don't significantly contribute to the overall shape. When I needed to change all the shapes' location uniformly, I could group the function and assign it a single name, "b1." Otherwise, I would have had to change the location of the shapes one by one, making the function lengthy and harder to understand, while also increasing the risk of bugs in the code.
Custom Properties (dx, dy) (NEW)	31 - 43	Custom properties are properties created by the programmer to store data inside an object.	I used the newly learned dx and dy to control the change of my 2 birds into a wrap-around motion, mimicking their flight in the air. With dx and dy, the change was much easier to code, and the flight's slope was clearly marked and simple to control.

Conditionals (if Statements)	94 - 130	An if statement indicates that the program will execute its code only if the condition is met.	I used a lot of if statements to test and push forward the game. For instance, to change the label on the scene, I would test whether it had met the conditions for a change. I would use something like “if curtain.y2 <= 70” to push forward with the logic and make “firstLabel.visible = False” and “secondLabel.visible = True.” Logic like this is used all throughout my code to allow certain things to happen when I want them to happen.
Event Functions (NEW)	140 - 143	An event function is a function that runs when a certain goal is achieved. For “onStep”, the code runs automatically on 30 steps per second by default.	2 event functions were used: “onMousePress” and “onStep.” For “onMousePress”, the action is performed only when the user presses the screen. Using this function, I was able to interact with the user, such as letting him open the curtains or shoot the birds by clicking on the screen. On the other hand, “onStep” is something I learned within the unit, and it can allow things to move without the user doing anything; they create animations such as the birds flying in the sky, the river flowing downstream, the sunlight, and the pot boiling food. Adding “onStep” made my game seem more visually appealing and logical. I also have to switch the automatic “app.stepsPerSecond” from 30 to 15 to ensure my code is comfortable and not too fast for the user.
Shape Methods	130	Shape methods are built-in functions that check on shape objects and control the spe to perform actions, like “shape.hits(x, y)” or “toFront()”.	I used shape methods to test the if-statements, the most common one I used is “light. contains(mouseX, mouseY)”, which makes sure that the user pressed on a specific object we want them to press on. In my program, I checked if the user pressed on my birds and the light. When they are pressed, the conditions below the if statement are performed.

Arithmetic Operators	164 - 167	Arithmetic operators are used for calculations in the program. Common operators used include =, +=,==,+, //, and %.	I used arithmetic operators to achieve the change in location, opacity, and values. For instance, the value of the “number of birds” and “number of rabbits” increases every time a bird or rabbit is caught; therefore, I had to use += 1 to change the value of their label. On the other hand, opacity could also be changed, for instance, the black background’s opacity gradually increases by 5 through “newBG.opacity += 5.” Overall, Arithmetic Operators allow me to “animate” things a bit more, though adding or subtracting some visual appearance.

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

Something I would absolutely change is the ending of the story. Now, the program performs that the pot will automatically boil once the lights are turned on, and it changes into a dark red soup. After the soup is boiled, the code ends with the text “dinner is finally ready.” I feel like this design is too lame, and it is programmed only because I had no more time left to make a better design. If I had more time, I would like to make the cooking part more enjoyable and interactive. I would like for the user to “put the supplies into the pot” themselves, and they can open the fire to cook. After the soup is finished, the user can put it onto a plate and eat it by clicking on the screen. After this, the lights gradually darken, indicating that the day is finally over. On the other hand, If I had the chance to start on scratch, I would also like to reduce the amount of interactions in the outdoor section, maybe by deleting the “open curtian” interaction and spend more time after the user gets home, then this game will seem much smoother and the time spent on indoors and outdoors will be more balance and seem more like a game. But right now, I would like to add a mechanism that, after you press it, the curtains go up, so it does not feature the user randomly clicking on the screen in order to open the curtains, which is not realistic.

I mostly stuck to my plan but made slight alterations, especially towards the end of the grame, initally, I wanted a “interactive cooking session” with the user; however, it was then changed to a lame animation for the user to watch. I also changed the outdoor to indoor transition, something I am pretty proud of. The changes were due to the lack of time, and were not exactly beneficial to the program, but within a limited amount of time, I am still very satisfied with what I produced. If I had another chance, I would change my time organization; otherwise, I like the clarity of my code and the aesthetic design of the program.

I asked my teacher for help a lot, especially when I meet bugs that are challenging for me to debug. On the other hand, my friend, Coco, also provided some supportive advice on my code and some suggestions on how the game will go. I also utilized the past assignment I finished and the lessons on CMU CS as a reference.

Time management is absolutely something I wish I had changed; with better and more organized time management, I wouldn’t be short on time for coding the indoor section. Therefore, if I had another chance, I would absolutely alter my time planning and aim to be more efficient so I can finish both parts smoothly and ideally. On the other hand, I still want to point out that coming to the front section in the given time was already a challenge for me, so I should still be proud of myself:)